Supplementary Material for SG-ALNS: Segment-Level Graph RL for ALNS Configuration in the VRPTW

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Implementation Details

Instances and state graph. With the depot horizon $H = l_0$, the six static node features are $x_i/100$, $y_i/100$, d_i/Q, e_i/H, l_i/H and σ_i/H. The six dynamic features are computed on the *current* solution x (not on the best one): waiting time, slack $l_i - b_i$ and forward time slack (Savelsbergh, 1992), each divided by H; cumulative load divided by Q; relative position in the route; and a routed flag (the depot has zeros). The edge set contains, for every customer, directed arcs to its 10 nearest customers, arcs between the depot and its 10 nearest customers, and both directions of every arc of the current routes (depot legs included). Each edge carries $c_{ij}/c_{\max}$ and a flag (0: neighbour arc, 1: route arc). The graph therefore grows with n, and mean and max pooling give an embedding of fixed size. The global vector holds t/T, τ/τ_0, the number of iterations since the last new best divided by $0.2T$ (capped at 1), $(f(x) - f(x^\star))/f(x^\star)$, $(f(x_0) - f(x^\star))/f(x_0)$ with x_0 the solution after vehicle reduction, the number of routes of x divided by that of the I1 solution, and the success rate (share of uses that improved the current solution) of each of the nine operators in the last segment. SG-ALNS-C appends $n/200$ and $T/5{,}000$.

Roulette wheel. After each segment, $w_o \leftarrow \max\{(1-\rho)w_o + \rho\,\psi_o/u_o,\ 10^{-3}\}$, with $\psi_o/u_o := 0$ for an operator that was not used, so the weight of an unused operator decays geometrically towards the floor. Destroy and repair probabilities are the weights normalised within each group. Scores per use are 33 (new best), 9 (better than the current solution) and 13 (accepted worse solution). A repair that cannot insert every customer is rejected and scores 0.

Factorised policy. The three heads are categorical distributions over a^1 (19), a^2 (5) and a^3 (3) computed from the same embedding, i.e. conditionally independent given the state: $\log \pi(a \mid s) = \sum_k \log \pi_k(a^k \mid s)$, and the entropy used in the PPO loss is the sum of the three head entropies (exact for this factorisation). In an ablation, the removed head is masked to its expert value and contributes zero to both. The output weights are scaled by 0.1 at initialisation and the biases of the expert values are set to 3 (a^1, RW) and 2 (a^2, default range; a^3, $m = 1$); the neutral-prior variant sets all biases to 0.

PPO. Rewards are divided by a running standard deviation; each update runs 4 epochs over minibatches of 256 decisions, with value coefficient 0.5 and gradient-norm clipping at 0.5. Every 5 updates the greedy policy is scored against ALNS-RW on 24 fixed generated instances ($n = 100$); this validation is only logged, and the *final* checkpoint is evaluated. Training instances, validation instances (seed 12345) and the 24 tuning instances of the tuned baselines (seeds 90000–90023) are drawn from disjoint random streams of Solomon's generator; no benchmark instance is used for any selection.

Objective and reward. The best solution $x^\star$ is always tracked with $f = 10^4\mathrm{NV} + \mathrm{TD}$. Over all 17,772 runs of this study, the largest distance difference between two solutions of the same instance is 2,569 (the largest TD is 11,354), so f orders every pair of our solutions lexicographically (NV first), also at $n = 400$. The reward of Eq. (1) is a learning signal only: one saved vehicle is worth $\lambda = 5$ points, i.e. as much as a 5% reduction of the initial distance. Unlike f, the reward therefore does not enforce the vehicle-first order: it is a shaping choice, and Table S15 varies λ.

TABLE S1
SOLOMON BENCHMARK ($n = 100$, $T = 5{,}000$, 5 SEEDS)

Class	BKS		ALNS-Uni		ALNS-RW		SG-ALNS		Tuned static	
	CNV	CTD	CNV	CTD	CNV	CTD	CNV	CTD	CNV	CTD
C1	90	7,455	**90.0**	**7,455**	**90.0**	**7,455**	**90.0**	**7,455**	**90.0**	**7,455**
C2	24	4,719	**24.0**	4,771	**24.0**	4,769	**24.0**	**4,719**	**24.0**	4,769
R1	143	14,524	149.8	14,496	149.6	14,416	148.2	14,530	**148.0**	**14,549**
R2	30	10,461	33.2	10,469	33.4	10,396	**33.0**	**10,439**	33.4	10,406
RC1	92	11,073	97.0	11,051	97.4	10,978	**95.8**	**11,005**	96.2	10,970
RC2	26	8,954	**27.0**	9,110	**27.0**	9,100	**27.0**	**9,040**	**27.0**	9,041
Total	405	57,187	421.0	57,352	421.4	57,115	**418.0**	**57,188**	418.6	57,190
Gap TD (%)[a]	–		2.09		1.86		**1.48**		1.69	
Primal integral (%)	–		5.44		5.58		**5.05**		5.07	
Time (s)	–		1.10		1.09		2.15		0.97	

[a] mean over runs whose NV equals the reference NV.

CNV/CTD: vehicles and distance summed over the instances of a class (mean over seeds). Bold: best among the compared methods (lexicographically: CNV, then CTD).

TABLE S2
GREEDY ACTIONS OF THE TRAINED POLICIES ON SOLOMON ($T = 5{,}000$

Model	a^1 (operators)	a^2 (q)	a^3 (m)	Share	#
SG-ALNS	RW weights	30%	3	1.00	1
w/o expert	Worst/Regret-3	30%	3	1.00	1
w/o a^1	RW weights	20%	1	0.73	2
w/o a^2	Worst/Regret-3	$\mathcal{U}[5, 30]\%$	3	1.00	1
w/o a^3	Worst/Regret-3	30%	1	1.00	1
w/o dyn. TW	Worst/Regret-3	30%	0.3	1.00	1
MLP encoder	Worst/Regret-3	30%	0.3	1.00	1
SG-ALNS, seed 1	Worst/Regret-3	30%	0.3	1.00	1
SG-ALNS, seed 2	Wait/Regret-3	30%	1	1.00	1
SG-ALNS-C	Worst/Regret-3	$\mathcal{U}[5, 30]\%$	0.3	1.00	1
Iteration-level GNN	Worst/Regret-3	$\mathcal{U}[5, 30]\%$	1	1.00	1
Iteration-level MLP	Worst/Regret-3	$\mathcal{U}[5, 30]\%$	1	1.00	1

All decisions of all runs). *Share*: fraction of decisions equal to the most frequent action; #: number of distinct actions observed.

TABLE S3
TRAINING SEEDS OF SG-ALNS (SOLOMON, $T = 5{,}000$)

Seed	CNV	Gap %	PI %	vs. ALNS-RW	Most frequent action	Share
0	418.0	1.48	5.05	-3.4 (0.012)	RW weights, q=30%, m=3	1.00
1	420.2	2.05	5.24	-1.2 (0.136)	Worst/Regret-3, q=30%, m=0.3	1.00
2	418.8	2.11	4.94	-2.6 (0.041)	Wait/Regret-3, q=30%, m=1	1.00
3	419.2	1.57	5.15	-2.2 (0.058)	Worst/Regret-3, q=30%, m=1	0.60
4	420.2	1.64	5.20	-1.2 (0.206)	Worst/Regret-3, q=30%, m=1	1.00

5 evaluation seeds. vs. ALNS-RW: ΔCNV and two-sided Wilcoxon p on per-instance vehicles; last columns: greedy action chosen most often and its share.

S0 TABULAR RESULTS

Statistics. The paper bases every conclusion on hierarchical, two-sided tests (fleet first, then distance on equal-fleet instances) with TOST equivalence (Table IV of the paper). Tables S7, S9 and S10 additionally list W/T/L on f with one-sided Wilcoxon p-values; these are descriptive and are not used for any conclusion.

Tables S1, S2 and S3 give the class-wise Solomon results, the greedy actions of all policies and the training-seed study that are shown graphically in Fig. 3 of the paper and Fig. S3.

S1 HEADROOM AND STATIC CONFIGURATIONS

Table S4 compares static choices with ALNS-RW on generated instances before training: no static choice dominates ALNS-RW, which motivates the expert-inclusive action space. Table S5 lists the static-configuration grid used to select the tuned static baseline, and Table S6 the dynamic-headroom study (two-phase schedules).

TABLE S4
HEADROOM ANALYSIS

Configuration	ΔNV	ΔTD (%)
ALNS-RW (reference)	+0	+0.00
Lead pair String/Regret-3	-6	+0.09
RW operators, fixed $q = 30\%$	-3	+0.12
RW operators, fixed $q = 20\%$	-3	+0.13
Lead pair Shaw/Regret-3	-2	+0.23
Lead pair Worst/Regret-3	-2	+0.34
RW operators, fixed $q = 10\%$	+4	+1.41
RW operators, fixed $q = 5\%$	+11	+2.17
Lead pair Route/Noise	+6	+2.28
Lead pair Route/Greedy	+0	+3.01

Static configurations vs. ALNS-RW on 24 generated instances ($n = 100$, $T = 2{,}500$, 3 seeds). ΔNV: vehicles summed over 72 runs; ΔTD: mean relative distance difference (positive = worse than RW).

TABLE S5
STATIC-CONFIGURATION GRID (30 CONFIGURATIONS) ON 24 GENERATED TUNING INSTANCES ($n = 100$, $T = 5{,}000$, 3 SEEDS)

Static configuration	Score	ΔNV
String/Regret-3, q=30%, m=0.3†	+0.70	-8
String/Regret-3, q=30%, m=1	+0.63	-5
String/Regret-3, q=20%, m=1	+0.55	-4
String/Regret-3, q=$\mathcal{U}[5, 30]$%, m=1	+0.49	-8
RW weights, q=30%, m=0.3	+0.47	-2
...		
RW weights, q=30%, m=3‡	+0.36	-5
ALNS-RW (reference)	+0.00	+0
RW weights, q=5%, m=0.3	-2.88	+10
RW weights, q=5%, m=3	-3.06	+8

No benchmark instance is used. Score: $5\Delta\text{NV} + 100\Delta\text{TD}/\text{TD}$ w.r.t. ALNS-RW (higher is better), i.e. the reward weighting. †: selected as *Tuned static*; ‡: the constant action SG-ALNS converged to.

TABLE S6
DYNAMIC HEADROOM

Schedule	Score
String/Regret-3, 20%, m=0.3 $\rightarrow$ String/Regret-3, 30%, m=3	+0.81
String/Regret-3, 30%, m=0.3 $\rightarrow$ RW weights, 30%, m=1	+0.81
String/Regret-3, 20%, m=0.3 $\rightarrow$ RW weights, 30%, m=3	+0.79
String/Regret-3, 20%, m=0.3 $\rightarrow$ String/Regret-3, 30%, m=1	+0.79
String/Regret-3, 20%, m=1 $\rightarrow$ RW weights, 30%, m=1	+0.78
String/Regret-3, 30%, m=0.3 (constant)	+0.70
String/Regret-3, 30%, m=1 (constant)	+0.63
String/Regret-3, 20%, m=1 (constant)	+0.55

Best two-phase schedules (first half $\rightarrow$ second half of the main phase) and best constants among the 10 best static actions, on the 24 generated tuning instances (3 seeds, $T = 5{,}000$). Score as in Table S5.

S2 ABLATIONS

Each ablation is a separately trained policy (seed 0). Because every greedy policy is (almost) state-independent (Fig. 3a of the paper), the differences reflect the constant configuration each training run converged to, not the value of the removed component.

S3 RUN-TIME BREAKDOWN AND CONVERGENCE

Table S8 splits the run time into the vehicle-reduction phase, the ALNS main phase and the policy overhead (features and inference). The policy accounts for 3–4% of the run time at $n = 400$; the rest is ALNS time, which depends on the operators and temperature selected. Figure S1 shows convergence on Solomon, and Fig. S2 the effect of the budget.

TABLE S7
ABLATIONS ON SOLOMON ($T = 5{,}000$, 5 SEEDS), EACH A SEPARATELY TRAINED POLICY (SAME BUDGET AND SEED)

Method	CNV	CTD	Gap (%)	PI (%)	Time (s)	W/T/L	p_{Holm}
SG-ALNS	418.0	57,188	1.48	5.05	2.15	–	–
w/o expert actions & prior	419.0	57,017	1.40	5.20	2.47	18/16/22	0.657
w/o a^1 (RW operators)	421.6	57,142	1.85	5.65	1.37	27/16/13	0.002
w/o a^2 (default q range)	420.6	57,269	1.95	5.32	1.54	26/16/14	0.002
w/o a^3 (scheduled τ)	420.2	57,077	1.64	5.20	2.46	23/16/17	0.062
w/o dynamic TW features	420.2	57,281	2.05	5.24	2.46	28/16/12	0.002
MLP encoder (no message passing)	420.2	57,281	2.05	5.24	2.35	28/16/12	0.002

Columns as in Table III. Because every greedy policy is state-independent (Table S2), these differences reflect the constant configuration each training run converged to.

TABLE S8
WALL-CLOCK BREAKDOWN PER RUN (SECONDS)

Set	Method	T	Reduction	ALNS	Policy	Total
Solomon	ALNS-RW	10,000	0.13	1.91	0.00	2.04
Solomon	Tuned ALNS-RW	10,000	0.12	2.59	0.00	2.71
Solomon	Tuned static	10,000	0.12	2.40	0.00	2.57
HG-400	ALNS-RW	5,000	4.86	12.69	0.00	17.55
HG-400	Tuned ALNS-RW	5,000	4.77	17.66	0.00	22.42
HG-400	Tuned static	5,000	4.80	16.09	0.00	21.34
HG-400	SG-ALNS	5,000	4.68	23.26	0.96	29.30
HG-400	SG-ALNS-C	5,000	4.69	17.77	0.91	23.76

Vehicle-reduction phase, ALNS main phase, and policy overhead (feature construction and network inference).

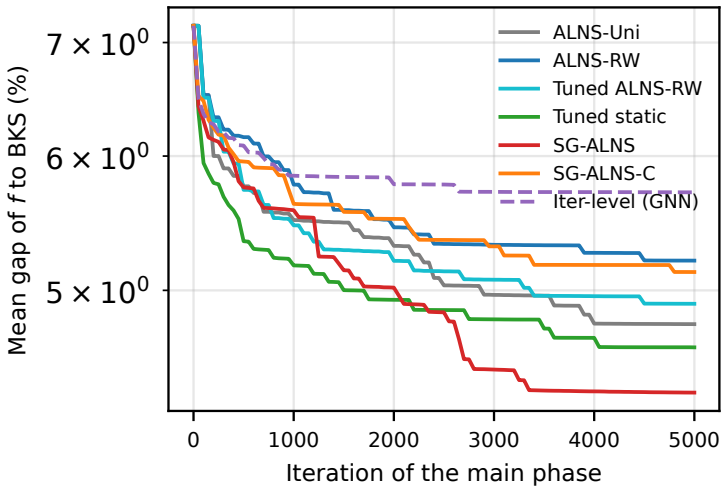


Fig. S1. Mean gap of the best-so-far objective f to the BKS along the main phase (Solomon, 3 seeds, log scale).

S4 TIME-MATCHED COMPARISON AND BUDGETS

S5 ACTIONS OF SG-ALNS-C

S6 INSTANCE SIZE

S7 LEARNED ACTIONS

Table S13 lists the greedy action of every trained policy; the six controls are summarised in Table I of the paper.

S8 BKS CROSS-CHECK AND REPRODUCIBILITY

The Solomon BKS table was cross-checked against the 56 CVRPLIB solutions (DIMACS convention: distance-only objective with distances truncated to one decimal). Re-evaluated with unrounded distances, 48 of them are feasible; on the 19 instances where they use the BKS fleet, they reproduce the BKS distance exactly. Over 8,232 Solomon runs, no solution dominates a

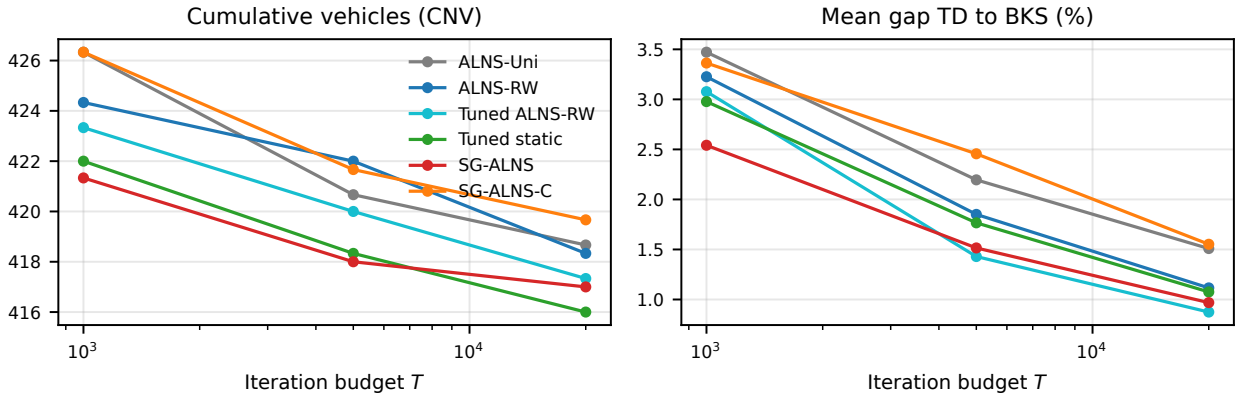


Fig. S2. Cumulative vehicles and TD gap on Solomon as a function of the budget T (3 seeds).

TABLE S9
TIME-MATCHED COMPARISON ON SOLOMON (5 SEEDS)

Method	T	Time (s)	CNV	CTD	Gap (%)
SG-ALNS	5,000	2.15	418.0	57,188	1.48
SG-ALNS-C	5,000	1.71	422.0	57,512	2.44
ALNS-RW	10,000	2.04	419.8	56,987	1.40
Tuned ALNS-RW	10,000	2.71	418.8	56,931	1.25
Tuned static	10,000	2.57	417.8	56,990	1.28

The baselines receive twice the iterations, which roughly equalises wall-clock time with SG-ALNS at $T = 5{,}000$.

TABLE S10
ITERATION BUDGET ON SOLOMON (3 SEEDS)

	CNV					Gap TD (%)					SG-ALNS vs.	
T	RW	RWt	St	SG	SG-C	RW	RWt	St	SG	SG-C	RWt	St
1,000	424.3	423.3	422.0	421.3	426.3	3.23	3.08	2.98	2.54	3.36	25/14/17 (0.002)	27/14/15 (0.028)
5,000	422.0	420.0	418.3	418.0	421.7	1.85	1.43	1.77	1.52	2.46	24/17/15 (0.064)	21/16/19 (0.150)
20,000	418.3	417.3	416.0	417.0	419.7	1.12	0.87	1.08	0.97	1.55	21/17/18 (0.506)	16/16/24 (0.753)

CNV and gap for ALNS-RW, tuned ALNS-RW (RWt), tuned static (St), SG-ALNS (SG) and SG-ALNS-C (SG-C); last columns: SG-ALNS vs. tuned ALNS-RW and vs. tuned static (W/T/L, one-sided Wilcoxon p). Tuned baselines were tuned for $T = 5{,}000$; SG-ALNS was trained with $T = 2{,}500$ and SG-ALNS-C with $T \in \{1{,}000, 2{,}500, 5{,}000\}$.

BKS, and 25 BKS are reproduced exactly. Gehring–Homberger instances with $n = 400$ were converted from the VRPLIB files of the PyVRP instance repository; the same converter reproduces our $n = 200$ files exactly. The script `run_all.sh` regenerates every result, and `make_paper.sh` every number, table and figure of the paper.

S9 ADDITIONAL ANALYSES

The state-sensitivity probe of the trained policies is Table V of the paper. *How much tuning is needed?* Table S14 selects constant actions on half of the generated tuning instances and scores them on the other half. *Reward.* Table S15 retrains seed 0 with other vehicle weights λ. *Iteration-level agents near their horizon.* Table S16 evaluates them at $T = 200$. *Instance classes at* $n = 400$. Table S17 splits the $n = 400$ results by class.

S10 ADDITIONAL FIGURES

TABLE S11
GREEDY ACTIONS OF SG-ALNS-C BY INSTANCE SIZE n AND BUDGET T

n	T	Mode, first half	Mode, second half	Share	#	$\bar{q}$ (%)
100	1,000	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	1.00	1	17.5
100	5,000	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	1.00	1	17.5
100	20,000	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	1.00	1	17.5
200	5,000	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	1.00	1	17.5
400	5,000	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3	1.00	1	17.5

Most frequent action in the first and second half of the run, share of the most frequent action over the run, number of distinct actions, and mean destroy fraction ($\mathcal{U}[5,30]\%$ counted as 17.5%).

TABLE S12
GEHRING–HOMBERGER INSTANCES WITH $n = 200$ AND $n = 400$ ($T = 5{,}000$, 3 SEEDS, 60 INSTANCES EACH)

n	Method	CNV	CTD	Gap (%)	PI (%)	Time (s)	vs. ALNS-RW
200	ALNS-RW	697.7	173,654	1.93	0.64	2.92	–
	Tuned ALNS-RW	697.7	173,225	1.66	0.62	6.32	34/6/20 (0.012)
	Tuned static	697.7	173,741	1.93	0.53	3.43	32/4/24 (0.120)
	SG-ALNS	697.3	174,722	2.41	0.65	5.28	18/6/36 (0.979)
	SG-ALNS-C	698.0	172,937	1.66	0.56	6.97	34/6/20 (0.016)
400	ALNS-RW	1397.3	411,122	2.79	0.66	17.55	–
	Tuned ALNS-RW	1397.0	411,267	2.75	0.63	22.42	35/0/25 (0.327)
	Tuned static	1395.7	417,026	3.79	0.47	21.34	27/1/32 (0.844)
	SG-ALNS	1396.0	422,567	5.01	0.61	29.30	17/1/42 (1.000)
	SG-ALNS-C	1396.7	405,283	1.53	0.51	23.76	43/0/17 (<0.001)

All policies and configurations were obtained with $n \leq 200$ (SG-ALNS-C) or $n \leq 100$ (all others). Gap and PI relative to the best solution found in the experiment; last column: W/T/L vs. ALNS-RW (one-sided Wilcoxon).

TABLE S13
GREEDY ACTIONS OF THE TRAINED POLICIES ON SOLOMON

Policy	Most frequent action	Share	#	Switch./run	CNV
PPO, seed 0	RW weights, q=30%, m=3	1.00	1	0.0	418.0
PPO, seed 1	Worst/Regret-3, q=30%, m=0.3	1.00	1	0.0	420.2
PPO, seed 2	Wait/Regret-3, q=30%, m=1	1.00	1	0.0	418.8
PPO, seed 3	Worst/Regret-3, q=30%, m=1	0.60	2	6.6	419.2
PPO, seed 4	Worst/Regret-3, q=30%, m=1	1.00	1	0.0	420.2
PPO, SG-ALNS-C	Worst/Regret-3, q=$\mathcal{U}$, m=0.3	1.00	1	0.0	422.0
PPO, neutral prior, seed 0	Worst/Regret-3, q=$\mathcal{U}$, m=0.3	0.78	2	1.7	422.6
PPO, w/o expert	Worst/Regret-3, q=30%, m=3	1.00	1	0.0	419.0
PPO, w/o a^1	RW weights, q=20%, m=1	0.73	2	8.7	421.6
PPO, w/o a^2	Worst/Regret-3, q=$\mathcal{U}$, m=3	1.00	1	0.0	420.6
PPO, w/o a^3	Worst/Regret-3, q=30%, m=1	1.00	1	0.0	420.2
PPO, w/o dyn. TW	Worst/Regret-3, q=30%, m=0.3	1.00	1	0.0	420.2
PPO, MLP encoder	Worst/Regret-3, q=30%, m=0.3	1.00	1	0.0	420.2
PPO, $\lambda = 1$	Worst/Regret-3, q=30%, m=0.3	1.00	1	0.0	420.2
PPO, $\lambda = 25$	Worst/Regret-3, q=30%, m=1	1.00	1	0.0	420.2
PPO, neutral prior, seed 1	Random/Regret-3, q=30%, m=0.3	1.00	1	0.0	420.6
PPO, neutral prior, seed 2	Worst/Regret-3, q=30%, m=0.3	1.00	1	0.0	420.2
PPO, seed 0, 5× training	Shaw/Regret-3, q=30%, m=0.3	1.00	1	0.0	419.6
Iteration-level GNN*	Worst/Regret-3, q=$\mathcal{U}$, m=1	1.00	1	–	–
Iteration-level MLP*	Worst/Regret-3, q=$\mathcal{U}$, m=1	1.00	1	–	–

$T = 5{,}000$, all decisions of all runs: most frequent action, its share, number of distinct actions, mean number of action switches per run, and CNV. *One decision per iteration, measured on 14 Solomon instances (`iter_actions.csv`).

TABLE S14
TUNING BUDGET VS. HELD-OUT QUALITY

b	CPU-min	Held-out score	P10	P90
1	1	-1.42	-3.18	+0.34
3	2	-0.44	-1.45	+0.68
10	5	+0.15	-0.57	+0.78
30	16	+0.47	+0.06	+0.83
100	53	+0.47	+0.18	+0.83
285	150	+0.18	+0.18	+0.18

285 constant actions, generated instances, $T = 5{,}000$, 3 seeds. For a budget of b random configurations, the best one on 12 selection instances is scored on 12 held-out instances (score vs. ALNS-RW: 5ΔNV$+100\Delta$TD/TD; mean, 10th and 90th percentile over 500 draws). CPU-min: cost of the selection runs.

TABLE S15
REWARD SENSITIVITY

λ	CNV	Gap %	vs. ALNS-RW	Most frequent action	Share
1	420.2	2.05	-1.2 (0.136)	Worst/Regret-3, q=30%, m=0.3	1.00
5 (paper)	418.0	1.48	-3.4 (0.012)	RW weights, q=30%, m=3	1.00
25	420.2	1.64	-1.2 (0.206)	Worst/Regret-3, q=30%, m=1	1.00

Seed 0, Solomon, $T = 5{,}000$, 5 seeds; λ: weight of a saved vehicle in Eq. (1). vs. ALNS-RW: ΔCNV and two-sided Wilcoxon p; last columns: most frequent greedy action and its share.

TABLE S16
SHORT BUDGET $T = 200$

Method	CNV	Gap %	Time s	vs. ALNS-RW
ALNS-RW	428.3	5.86	0.14	–
Tuned static	425.3	4.59	0.17	-3.0 (0.009)
SG-ALNS	425.0	5.37	0.20	-3.3 (0.026)
Iteration-level GNN	428.3	4.83	0.58	-0.0 (0.750)
Iteration-level MLP	428.3	4.83	0.38	-0.0 (0.750)

Solomon, 3 seeds; close to the 100-iteration training episodes of the iteration-level agents. vs. ALNS-RW: ΔCNV and two-sided Wilcoxon p.

TABLE S17
GEHRING–HOMBERGER $n = 400$ PER INSTANCE CLASS

Method	C1	C2	R1	R2	RC1	RC2
ALNS-RW	381.0 / 1.69	120.0 / 1.44	364.0 / 3.47	80.0 / 1.64	362.0 / 5.91	90.3 / 2.88
Tuned ALNS-RW	380.7 / 2.00	120.0 / 1.26	364.0 / 3.31	80.0 / 1.59	362.0 / 6.01	90.3 / 2.63
Tuned static	380.7 / 5.21	120.0 / 0.95	364.0 / 5.39	80.0 / 1.44	361.3 / 8.24	89.7 / 1.97
SG-ALNS	380.7 / 6.30	120.0 / 2.28	364.0 / 5.18	80.0 / 2.58	361.3 / 10.17	90.0 / 3.57
SG-ALNS-C	381.0 / 1.04	120.0 / 0.51	364.0 / 0.92	80.0 / 1.75	361.3 / 1.26	90.3 / 3.70

10 instances each, 3 seeds: cumulative vehicles / mean TD gap (%) to the best solution found, over runs with that fleet.

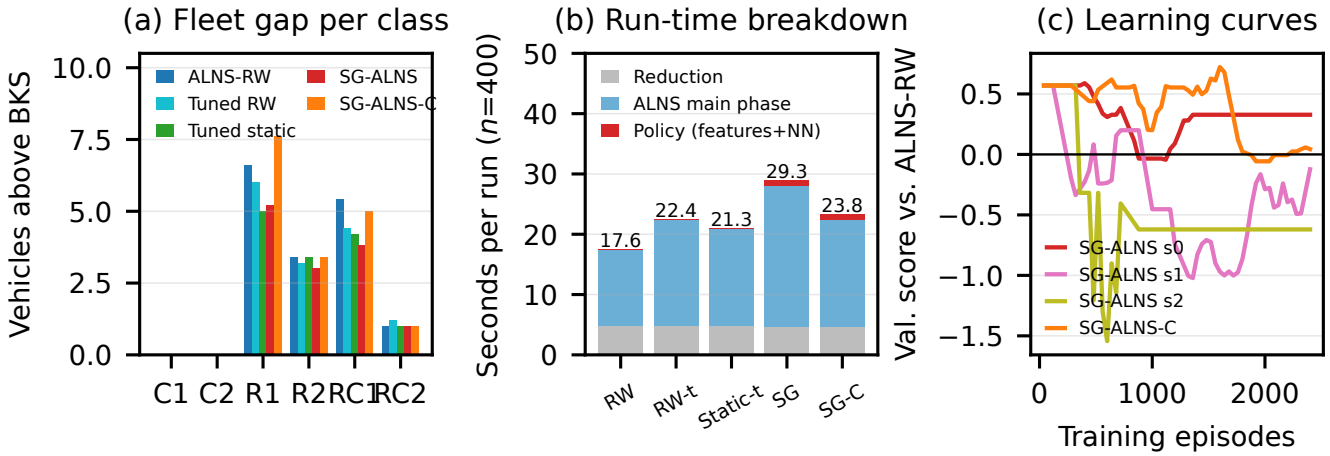


Fig. S3. (a) Vehicles above the BKS per Solomon class (sum over instances, mean over 5 seeds). (b) Mean run time per instance at $n = 400$, split into vehicle reduction, ALNS main phase and policy overhead (number: total). (c) Validation score against ALNS-RW on 24 generated instances during training (rolling mean of 4 evaluations).

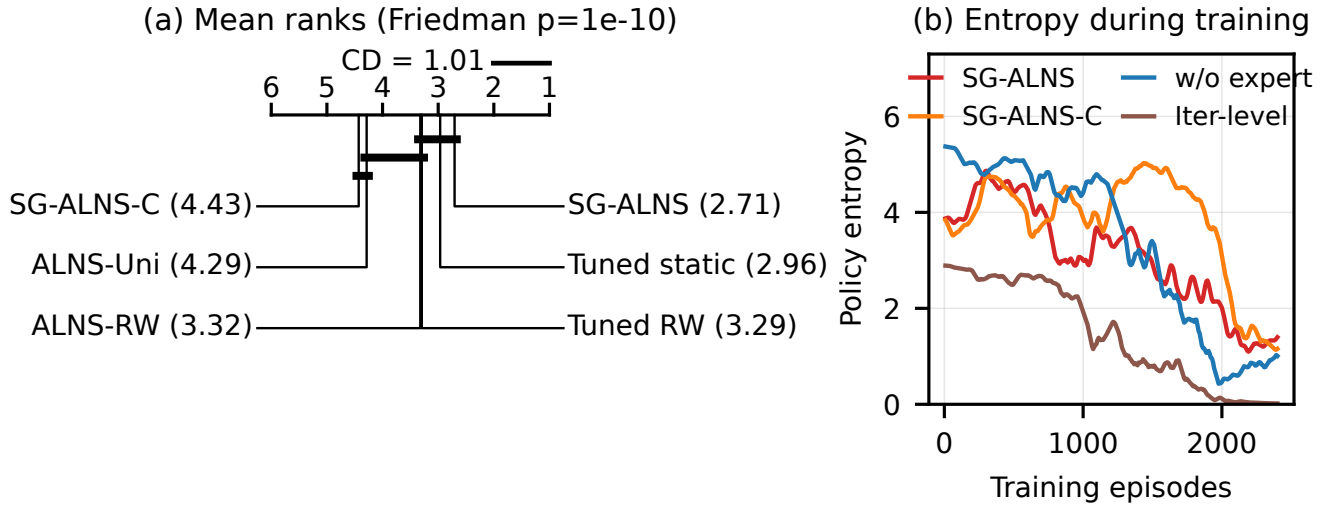


Fig. S4. (a) Critical-difference diagram (Friedman–Nemenyi, $\alpha = 0.05$) of the six methods run with 5 seeds on Solomon, $T = 5{,}000$; methods joined by a bar are not significantly different. (b) Policy entropy (sum over the three heads) during training.